

Geometry

Unit 10

Lesson 1 Homework

Name _____ Answer Key _____

Date _____

Period _____

Use the given context to answer questions 1-10.

Jamar orders an extra-large circular pizza that has an 18-inch (in.) diameter and is cut into 12 triangular slices.

1. What is the approximate area of an 18-inch pizza? Use 3.14 for π .

$$A = 3.14(9)^2 = 3.14(81) = 254.34 \text{ in}^2$$

2. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 6 slices?

$$\text{Fraction: } \frac{1}{2}$$

$$\text{Percentage: } 50\%$$

3. What is the area of 6 slices?

$$A = 127.17 \text{ in}^2$$

4. What is the central angle of 6 slices?

$$180^\circ$$

5. If the slices are cut precisely and equally, what fraction of the pizza is 8 slices?

$$\frac{8}{12} = \frac{2}{3}$$

6. What is the area of 8 slices?

$$A = \frac{2}{3}(254.34) = 169.56 \text{ in}^2$$

7. What is the central angle of 8 slices?

$$240^\circ$$

8. If the slices are cut precisely and equally, what fraction of the pizza is 1 slice?

$$\frac{1}{12}$$

9. What is the area of 1 slice?

$$A = \frac{1}{12}(254.34) = 21.195 \text{ in}^2$$

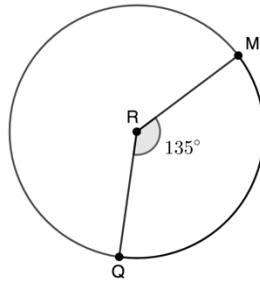
10. What is the central angle of 1 slice?

$$30^\circ$$

Geometry
Unit 10
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

For questions 1-2, Circle R is shown where points M and Q are on the circle.



1. What is the ratio of the central angle, $\angle QRM$, to the whole circle? Simplify the ratio.

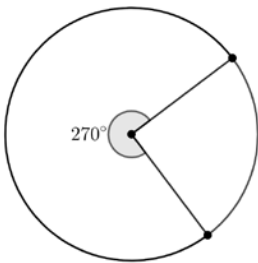
$$\frac{135}{360} = \frac{3}{8}$$

2. What fraction of the circle's circumference does $\widehat{MQ}$ represent?

$$\frac{225}{360} = \frac{5}{8} \text{ for the larger arc; } \frac{135}{360} = \frac{3}{8} \text{ for the smaller arc}$$

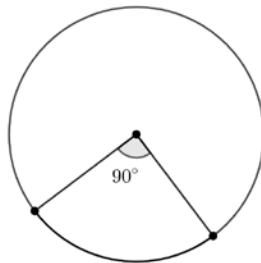
For questions 3-8, find the ratio of the arc length to the circumference of the circle.

3.



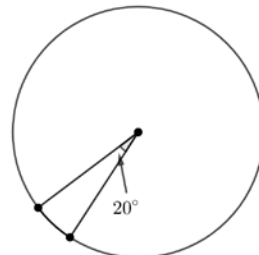
$$\frac{270}{360} = \frac{3}{4}$$

4.



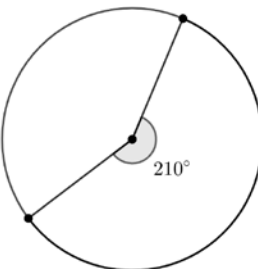
$$\frac{90}{360} = \frac{1}{4}$$

5.



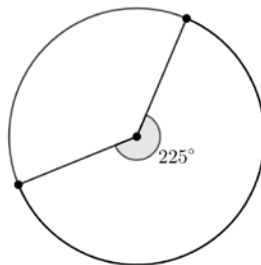
$$\frac{20}{360} = \frac{1}{18}$$

6.



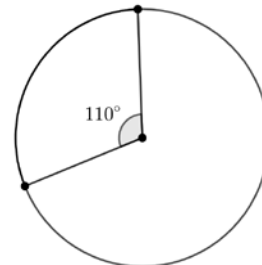
$$\frac{210}{360} = \frac{7}{12}$$

7.



$$\frac{225}{360} = \frac{5}{8}$$

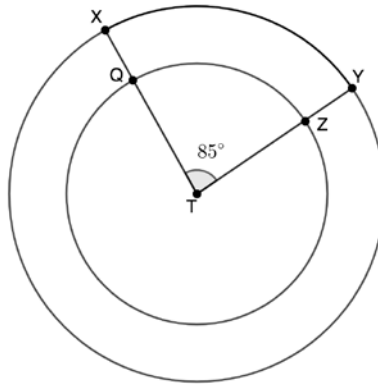
8.



$$\frac{110}{360} = \frac{11}{36}$$

For questions 9-10, consider the following information.

Points X and Y lie on Circle T . The radius $\overline{TY}$ measures 8 inches (in.). A smaller Circle T with a radius measure of 5 in. rests inside the larger Circle T , where points Q and Z lie on the smaller circle. The smaller Circle T has central angle $\angle QTZ$ that measures 85° and the larger Circle T has central angle $\angle XTY$ that measures 85° .



9. What is the ratio of the arc length to circumference of Circle T with radius measure of 8 in.?

$$\frac{85}{360} = \frac{17}{72}$$

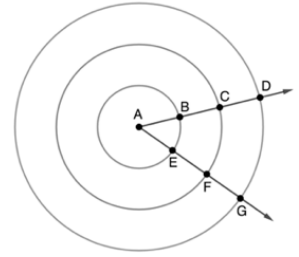
10. What is the ratio of the arc length to circumference of Circle T with radius measure of 5 in.?

$$\frac{85}{360} = \frac{17}{72}$$

Lesson 3 Homework

Period _____

and has radius $AD = 6$ units.



in terms of π .

	Circle	Circumference
1.	Innermost Circle	4π
2.	Middle Circle	8π
3.	Outer Circle	12π

Complete the table.

	Circle	Ratio of Circumference to Radius
4.	Innermost Circle	$\frac{4\pi}{2} = 2\pi$
5.	Middle Circle	$\frac{8\pi}{4} = 2\pi$
6.	Outer Circle	$\frac{12\pi}{6} = 2\pi$

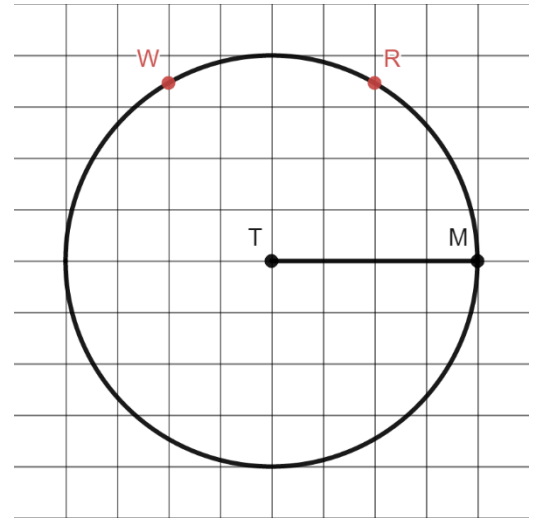
7. What would be the ratio of the circumference to the radius for a circle whose center is N and whose radius has a measurement of 9?

$$\frac{18\pi}{9} = 2\pi$$

Circle T is shown where $\overline{TM}$ is a radius.

8. Label point R on the circumference of the circle so that the resulting arc has an approximate length of 1 radian.

See grid.



9. Describe how you determined the location of point R .

Answers will vary. Example: Measured the length of the radius and measured the same length on the circle.

10. Label point W on the circumference of the circle so that the resulting arc has an approximate length of 2 radians.

See grid.

1. Circle B has a diameter of 14 inches (in.). Points H and W are on the circle and form the sector HBW . The central angle $\angle HBW$ in Circle B measures 150° . Determine the area of the sector. Express your answer in terms of π .

$$A = \frac{150}{360}(\pi)(7)^2 = 20.41\bar{6}\pi \text{ in}^2$$

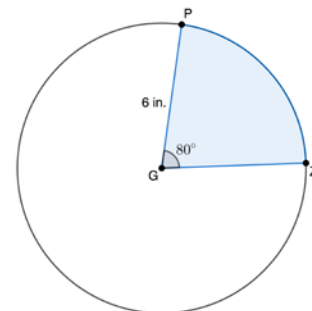
2. The area of a sector with a radius of 10 centimeters (cm) is 148.28 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

$$148.28 = \frac{x}{360}(\pi)(10)^2$$

$$170^\circ = x \text{ using } \pi = 3.14$$

3. Circle G has radius $\overline{GP}$ that measures 6 inches (in.) with central angle $\angle PGZ$, where $m\angle PGZ = 80^\circ$. Find the area of the shaded sector. Round your answer to the nearest hundredth.

$$A = \frac{80}{360}(\pi)(6)^2 = 25.12 \text{ in.}^2 \text{ using } \pi = 3.14$$



4. Circle H has a radius of 9 inches (in.). Points M and N are on the circle and form the sector MHN . The central angle, $\angle MHN$, in Circle H measures 115° . Determine the area of the sector. Express your answer in terms of π .

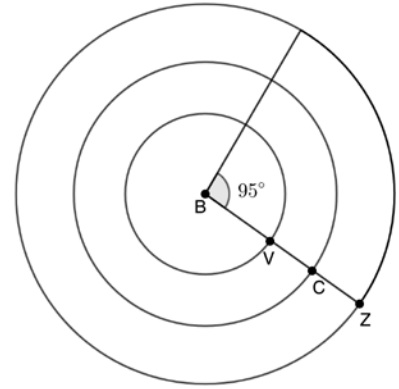
$$A = \frac{115}{360}(\pi)(9)^2 = 25.875\pi \text{ in.}^2$$

5. The area of a sector with a diameter of 12 feet (ft) is 62.8 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

$$62.8 = \frac{x}{360}(\pi)(6)^2$$

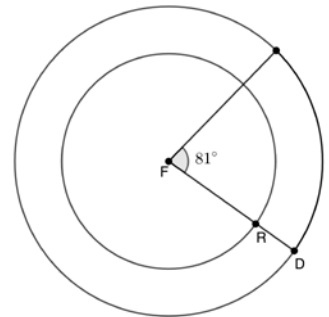
$$200^\circ = x \text{ using } \pi = 3.14$$

For questions 6-8, concentric circles with center B are shown with $m\angle B = 95^\circ$. The smallest Circle B has a radius $\overline{BV}$ that measures 12 meters (m). The largest Circle B has radius $\overline{BZ}$ and the middle Circle B has radius $\overline{BC}$. Each circle has a radius that is 12 m longer than the immediate prior circle.



6. What is the arc length of the circle with radius $\overline{BV}$?
 $\frac{95}{360}(2\pi)(12) \approx 19.89 \text{ m. using } \pi = 3.14$
7. What is the arc length of the circle with radius $\overline{BC}$?
 $\frac{95}{360}(2\pi)(24) \approx 39.77 \text{ m. using } \pi = 3.14$
8. What is the arc length of the circle with radius $\overline{BZ}$?
 $\frac{95}{360}(2\pi)(36) \approx 59.66 \text{ m. using } \pi = 3.14$

For questions 9-10, concentric circles with center F are shown with $m\angle F = 81^\circ$. The smallest Circle F has a radius $\overline{FR}$ that measures 15 meters (m) and the largest Circle F has radius $\overline{FD}$. The radius of the largest Circle F has a radius 7 m larger than the smaller Circle F .



9. What is the arc length of the circle with radius $\overline{FR}$?
 $\frac{81}{360}(2\pi)(15) \approx 21.195 \text{ m. using } \pi = 3.14$
10. What is the arc length of the circle with radius $\overline{FD}$?
 $\frac{81}{360}(2\pi)(22) \approx 31.086 \text{ m. using } \pi = 3.14$

1. Circle Y has points X , M , and R on the circle with $m\angle MYX = 37.2^\circ$. If $m\widehat{MR} = 146.8^\circ$ and $\widehat{MR}$ does not intersect $\widehat{MX}$, what is $m\angle RYX$?

$$m\angle RYX = 176^\circ$$

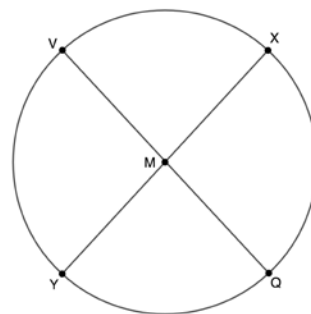
2. Circle G has radii $\overline{GH}$ and $\overline{GK}$. Find the value of x if $m\angle KGH = 122^\circ$ and $m\widehat{KH} = (5x + 7)^\circ$.

$$5x + 7 = 122$$

$$5x = 115$$

$$x = 23$$

For questions 3-4, Circle M has diameters $\overline{VQ}$ and $\overline{XY}$ that both intersect at point M .



3. Given $m\widehat{QY} = 80^\circ$, find the $m\widehat{VX}$.

$$m\widehat{VX} = 80^\circ$$

4. Given $m\widehat{QX} = 100^\circ$, find the $m\widehat{YV}$.

$$m\widehat{YV} = 100^\circ$$

For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(5x + 119)^\circ$ as the measure of the central angle, $\angle JQZ$. For the $m\widehat{ZJ}$, he uses the expression $(25x + 59)^\circ$.

5. What is the value of x ?

$$5x + 119 = 25x + 59$$

$$60 = 20x$$

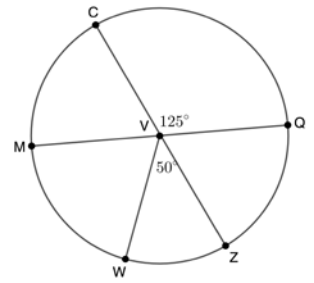
$$3 = x$$

6. What is the degree measure of the arc of the fan, $\widehat{ZJ}$?

$$134^\circ$$

7. In Circle V , $m\angle CVQ = 125^\circ$ and $m\angle ZVW = 50^\circ$. Calculate the $m\widehat{CQW}$, given that V is the center and $\overline{CZ}$ and $\overline{MQ}$ are diameters.

$$m\widehat{CQW} = 125 + 55 + 50 = 230^\circ$$

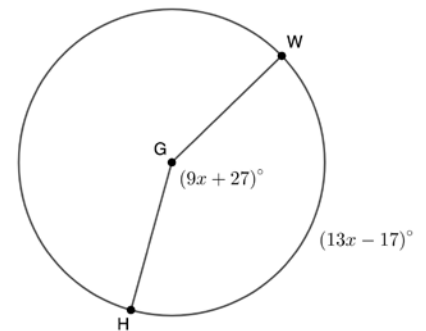


8. Circle Y has points X , Q , and W on the circle. The measure of the central angle, $\angle WYQ = 65^\circ$, $m\widehat{XW} = 117^\circ$, and $\widehat{XW}$ does not intersect $\widehat{WQ}$. Find $m\angle XYQ$.

$$360 - 65 - 117 = 178^\circ$$

For questions 9-10, consider the following context.

A horse is tied to a post at point G and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point W along the circumference of the circular area she can roam. The measure of the central angle, $\angle WGH$, is $(9x + 27)^\circ$, and the measure of the arc that the horse walked, $\widehat{WH}$, is $(13x - 17)^\circ$.



9. Find the value of x .

$$9x + 27 = 13x - 17$$

$$44 = 4x$$

$$11 = x$$

10. Find the measure of the arc that the horse walked, $\widehat{WH}$.

$$13(11) - 17 = 126^\circ$$

Geometry

Unit 10

Lesson 6 Homework

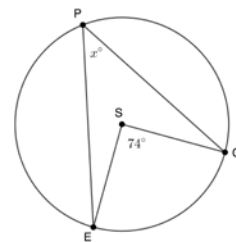
Name Answer Key

Date _____

Period _____

1. Circle S is shown where $\angle GSE$ is a central angle, $\angle GPE$ is inscribed in the circle, and $m\angle GSE = 74^\circ$. Find $m\angle GPE$.

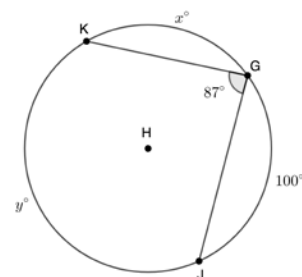
$$\frac{74}{2} = 37^\circ$$



For questions 2-3, Circle H has points K , G , and J on the circle. $\angle K G J$ is inscribed in Circle H measuring 87° , and the $m\widehat{GJ} = 100^\circ$.

2. Find the $m\widehat{KJ}$.

$$87 \cdot 2 = 174^\circ$$



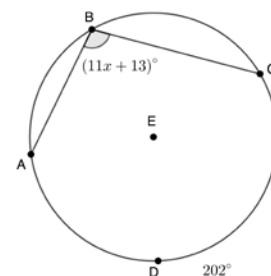
3. Find the $m\widehat{KG}$.

$$360 - 174 - 100 = 86^\circ$$

For questions 4-5, Points A , B , C , and D lie on Circle E . $\angle ABC$ is inscribed in the circle, the $m\angle ABC = (11x + 13)^\circ$, and the $m\widehat{CDA} = 202^\circ$.

4. Find the $m\angle ABC$.

$$\frac{202}{2} = 101^\circ$$



5. Find the value of x .

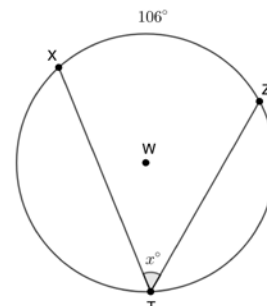
$$11x + 13 = 101$$

$$11x = 88$$

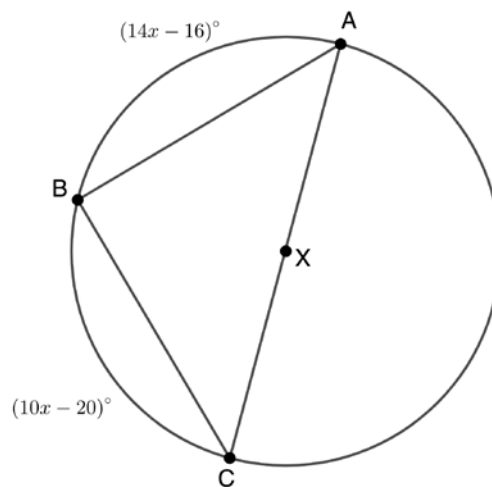
$$x = 8$$

6. Circle W is shown, where points X , Z , and T are on the circle. $\angle XTZ$ is inscribed in Circle W with $m\widehat{XZ} = 106^\circ$. Find the value of x .

$$\frac{106}{2} = 53$$



For questions 7-10, Circle X is shown with inscribed angle $\angle ABC$ and diameter $\overline{CA}$. The $m\widehat{AB} = (14x - 16)^\circ$ and the $m\widehat{BC} = (10x - 20)^\circ$.



7. Find the value of x .

$$14x - 16 + 10x - 20 = 180$$

$$24x - 36 = 180$$

$$24x = 216$$

$$x = 9$$

8. Find the $m\angle BAC$.

$$10(9) - 20 = 70$$

$$\frac{70}{2} = 35^\circ$$

9. Find the $m\angle ACB$.

$$14(9) - 16 = 110$$

$$\frac{110}{2} = 55^\circ$$

10. Find the $m\widehat{CB}$.

$$10(9) - 20 = 70^\circ$$

For questions 1-3, Circle Q is shown, where chords $\overline{RC}$ and $\overline{WX}$ intersect at point P , $m\widehat{CW} = 142^\circ$, $m\widehat{RX} = 82^\circ$, and $m\angle WPC = (15x - 23)^\circ$.

1. Find the $m\angle WPC$.

$$\frac{142+82}{2} = 112^\circ$$

2. Write a formula to find the value of x .

$$15x - 23 = 112$$

3. Find the value of x .

$$15x - 23 = 112$$

$$15x = 135$$

$$x = 9$$

4. Circle E is shown, where chords $\overline{BD}$ and $\overline{CA}$ intersect at point Z . Find the value of x for $\angle AZD$, given $m\widehat{CD} = 115^\circ$ and $m\widehat{AB} = 53^\circ$.

$$\frac{115+53}{2} = 84$$

$$x = 180 - 84 = 96$$

For questions 5-7, Circle F is shown, where chords $\overline{LZ}$ and $\overline{KW}$ intersect at point G , $m\widehat{LW} = 70^\circ$, $m\widehat{KZ} = (15x + 10)^\circ$, and $m\angle KGZ = 145^\circ$.

5. Write a formula to find the value of x .

$$\frac{15x+10+70}{2} = 145$$

6. Find the value of x .

$$\frac{15x+10+70}{2} = 145$$

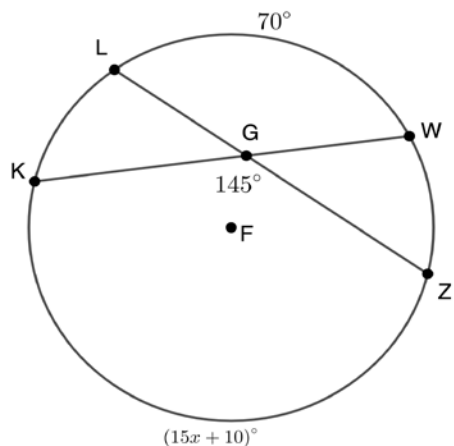
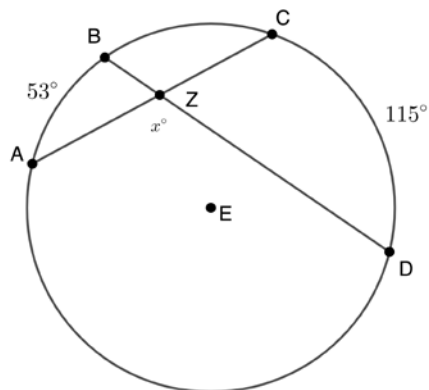
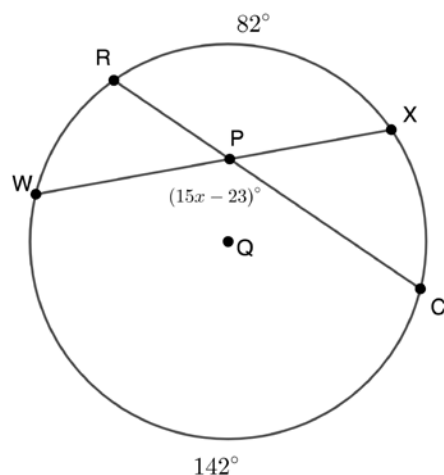
$$15x + 80 = 290$$

$$15x = 210$$

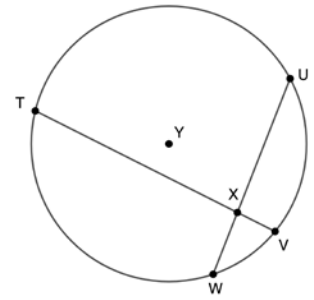
$$x = 14$$

7. Find the $m\widehat{KZ}$.

$$15(14) + 10 = 220^\circ$$



For questions 8-10, Circle Y is shown, where chords $\overline{VT}$ and $\overline{UW}$ intersect at point X .



8. If $m\widehat{TU} = 128^\circ$ and $m\widehat{VW} = 30^\circ$, what is the measure of $\angle TXU$?

$$\frac{128+30}{2} = 79^\circ$$

9. If $m\widehat{TW} = 152^\circ$ and $m\widehat{UV} = 40^\circ$, what is the measure of $\angle TXW$?

$$\frac{152+40}{2} = 96^\circ$$

10. If $m\widehat{VW} = 56^\circ$ and $m\angle TXU = 78^\circ$, what is the measure of $\widehat{TU}$?

$$78 = \frac{56+x}{2}$$

$$156 = 56 + x$$

$$100^\circ = x = \widehat{TU}$$

Geometry

Unit 10

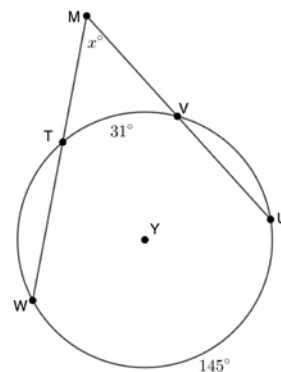
Lesson 8 Homework

Name Answer Key

Date _____

Period _____

For questions 1-2, Circle Y has points W, T, V , and U on the circle. Secant lines $\overline{WM}$ and $\overline{UM}$ intersect at point M outside the circle. The $m\widehat{UW} = 145^\circ$, $m\widehat{TV} = 31^\circ$, and $m\angle TMV = x^\circ$.



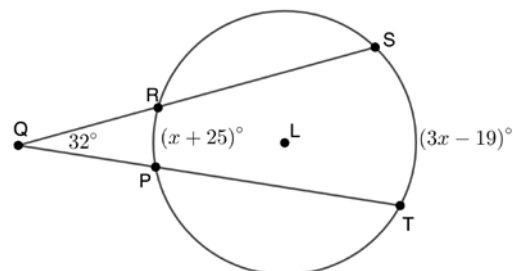
1. Write a formula that can be used to find the value of x .

$$x = \frac{145-31}{2}$$

2. Find the value of x .

$$x = 57$$

For questions 3-5, Circle L has points T, P, R , and S on the circle. Secant lines $\overline{TQ}$ and $\overline{SQ}$ intersect at point Q outside the circle. The $m\widehat{ST} = (3x - 19)^\circ$, $m\widehat{PR} = (x + 25)^\circ$, and $m\angle PQR = 32^\circ$.



3. Find the value of x .

$$\frac{(3x-19)-(x+25)}{2} = 32$$

$$2x - 44 = 64$$

$$2x = 108$$

$$x = 54$$

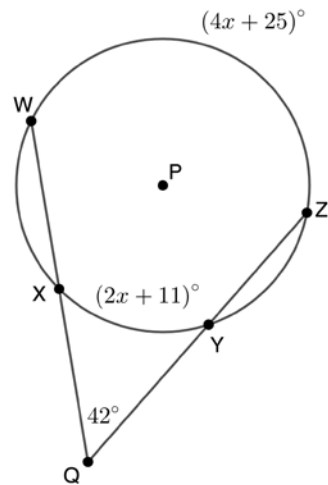
4. Find the $m\widehat{PR}$.

$$54 + 25 = m\widehat{PR} = 79^\circ$$

5. Find the $m\widehat{ST}$.

$$3(54) - 19 = m\widehat{ST} = 143^\circ$$

For questions 6-8, Circle P has points $W, X, Y,$ and Z on the circle. Secant lines $\overline{WQ}$ and $\overline{ZQ}$ intersect at point Q outside the circle. The $m\widehat{WZ} = (4x + 25)^\circ$, $m\widehat{XY} = (2x + 11)^\circ$, and $m\angle XQY = 42^\circ$.



6. Find the value of x .

$$\frac{(4x+25)-(2x+11)}{2} = 42$$

$$2x + 14 = 84$$

$$2x = 70$$

$$x = 35$$

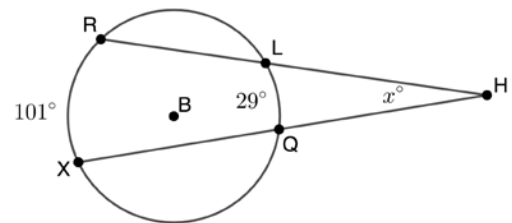
7. Find the $m\widehat{XY}$.

$$2(35) + 11 = m\widehat{XY} = 81^\circ$$

8. Find the $m\widehat{WZ}$.

$$4(35) + 25 = m\widehat{WZ} = 165^\circ$$

For questions 9-10, Circle B has points $R, L, Q,$ and X on the circle. Secant lines $\overline{XH}$ and $\overline{RH}$ intersect at point H outside the circle. The $m\widehat{RX} = 101^\circ$, $m\widehat{LQ} = 29^\circ$, and $m\angle LHQ = x^\circ$.



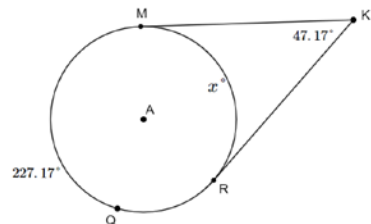
9. Write a formula that can be used to find the value of x .

$$x = \frac{101-29}{2}$$

10. Find the value of x .

$$x = 36$$

For questions 1-2, Circle A has points M , Q , and R on the circle. Tangents $\overline{MK}$ and $\overline{RK}$ intersect at point K outside the circle. The $m\widehat{MQR} = 227.17^\circ$, $m\widehat{MR} = x^\circ$, and $m\angle MKR = 47.17^\circ$.



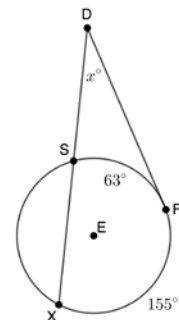
1. Write a formula that can be used to find the value of x .

$$x = 360 - 227.17 \text{ or } \frac{227.17 - x}{2} = 47.17$$

2. Find the value of x .

$$x = 132.83$$

For questions 3-4, Circle E has points X , S , and F on the circle. Tangent $\overline{FD}$ and secant $\overline{XD}$ intersect at point D outside the circle. The $m\widehat{XF} = 155^\circ$, $m\widehat{SF} = 63^\circ$, and $m\angle SDF = x^\circ$.



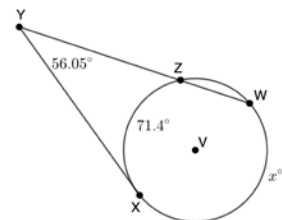
3. Write a formula that can be used to find the value of x .

$$x = \frac{155 - 63}{2}$$

4. Find the value of x .

$$x = 46$$

For questions 5-6, Circle V has points X , Z , and W on the circle. Tangent $\overline{XY}$ and secant $\overline{WY}$ intersect at point Y outside the circle. The $m\widehat{XW} = x^\circ$, $m\widehat{XZ} = 71.4^\circ$, and $m\angle XYZ = 56.05^\circ$.



5. Write a formula that can be used to find the value of x .

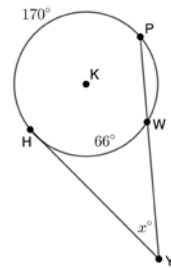
$$\frac{x - 71.4}{2} = 56.05$$

6. Find the value of x .

$$x = 183.5$$

7. Circle K has points P , W , and H on the circle. Tangent $\overline{HY}$ and secant $\overline{PY}$ intersect at point Y outside the circle. The $m\widehat{HP} = 170^\circ$, $m\widehat{HW} = 66^\circ$, and $m\angle HYW = x^\circ$. What is the $m\angle HYW$?

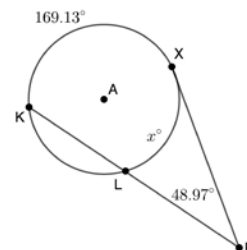
$$x = \frac{170 - 66}{2} = m\angle HYW = 52^\circ$$



8. Circle A has points X , L , and K on the circle. Tangent $\overline{XP}$ and secant $\overline{KP}$ intersect at point P outside the circle. The $m\widehat{KX} = 169.13^\circ$, $m\widehat{XL} = x^\circ$, and $m\angle XPL = 48.97^\circ$. What is the $m\widehat{XL}$?

$$\frac{169.13 - x}{2} = 48.97$$

$$x = m\widehat{XL} = 71.19^\circ$$



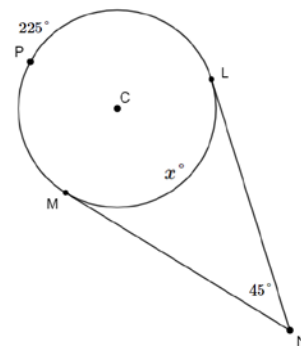
9. Circle C has points L , P , and M on the circle. Tangents $\overline{LN}$ and $\overline{MN}$ intersect at point N outside the circle. The $m\widehat{LPM} = 225^\circ$, $m\widehat{LM} = x^\circ$, and $m\angle LNM = 45^\circ$. What is the $m\widehat{LM}$?

$$x = 360 - 225 = m\widehat{LM} = 135^\circ$$

or

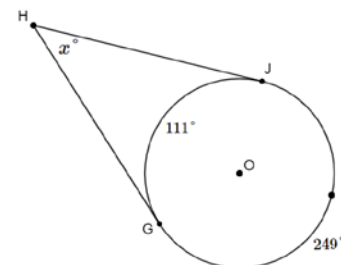
$$\frac{225 - x}{2} = 45$$

$$x = m\widehat{LM} = 135^\circ$$

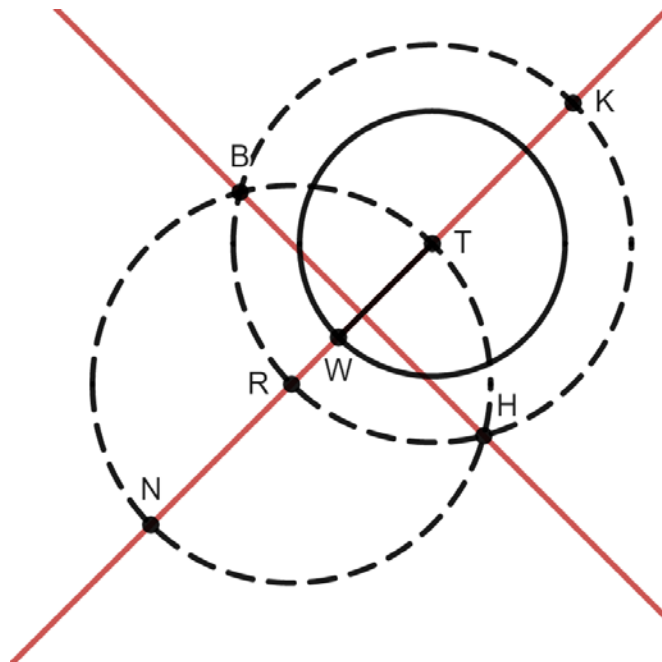


10. Circle O has points G , J , and K on the circle. Tangents $\overline{GH}$ and $\overline{JH}$ intersect at point H outside the circle. The $m\widehat{GK} = 249^\circ$, $m\widehat{GJ} = 111^\circ$, and $m\angle GHJ = x^\circ$. What is the $m\angle GHJ$?

$$x = \frac{249 - 111}{2} = m\angle GHJ = 69^\circ$$



Circle T is shown with radius $\overline{TW}$. The two dotted circles were constructed using a compass so that they are congruent to each other. Points K , H , R , and B are on the dotted Circle T and points B , H , T , and N on the dotted Circle R .



1. Select all of the line segments that are congruent to each other.

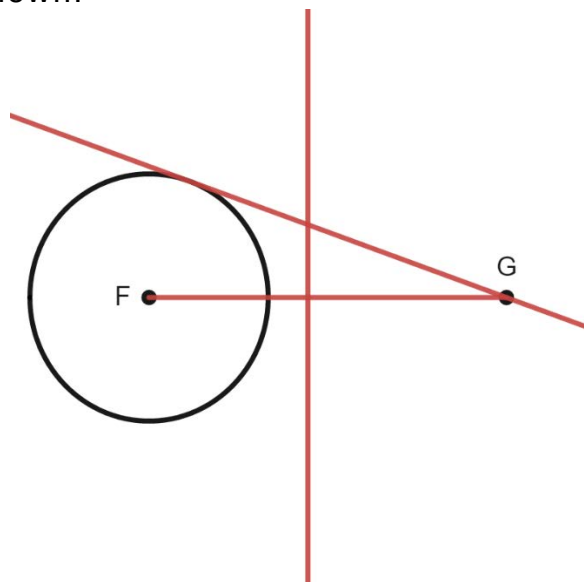
- ☐ $\overline{BK}$
☐ $\overline{BR}$
☒ $\overline{BT}$
☒ $\overline{TH}$
☒ $\overline{TK}$
☒ $\overline{TR}$
☐ $\overline{TW}$

2. Justify why the line segments chosen in question 1 are congruent.
 Answers may vary. Possible response: Line segments are all radii of Circle T .
3. Select all of the line segments that are congruent to each other.

- ☐ $\overline{BH}$
☐ $\overline{BN}$
☒ $\overline{BR}$
☐ $\overline{NH}$
☒ $\overline{NR}$
☒ $\overline{RH}$
☒ $\overline{RT}$
☐ $\overline{RW}$

4. Justify why the line segments chosen in question 3 are congruent.
 Answers may vary. Possible response: Line segments are all radii of Circle R .
5. Draw $\overline{NK}$ on the diagram.
 See diagram.
6. Draw $\overline{BH}$ on the diagram.
 See diagram.
7. Describe how $\overline{NK}$ and $\overline{BH}$ are related.
 They are perpendicular.
8. Justify whether $\overline{BH}$ is tangent to Circle T .
 Answers may vary. Possible response: $\overline{BH}$ is not tangent to Circle T because it intersects the circle twice instead of once.

Circle F and point G are shown.



9. Construct the perpendicular bisector of $\overline{FG}$.
See diagram.

10. Construct a line that is tangent to Circle F and goes through point G .
See diagram.